

THE SYNGAS STRATEGIST



DR. ROGER KUMAR

Founder and Managing Director of CASE Group

When furnace oil prices spike in the wake of a Middle East conflict, when LPG supply chains buckle under the weight of a European war, when natural gas contracts become instruments of geopolitical leverage, most Indian industries absorb the shock and wait. Dr. Roger Kumar built a company so they would not have to.

Since founding CASE Group in Faridabad in January 1993, the National Institute of Technology, Kurukshetra alumnus has pursued a single, disciplined thesis: that India's vast domestic coal reserves, properly converted through advanced gasification technology, can liberate its industrial base from the volatility of imported fuel markets. Thirty years later, with global energy geopolitics more turbulent than at any point in a generation, that thesis looks less like an engineering philosophy and more

like a national security argument.

Coal gasification is a thermo-chemical process that converts coal into syngas, a combustible mixture of carbon monoxide, hydrogen and methane. Rather than burning coal directly, CASE Group's gasifiers partially oxidise coal in a controlled, automated environment to produce a cleaner, more efficient fuel that can be fed directly into furnaces, kilns and dryers across steel plants, ceramic units, glass manufacturers, aluminium smelters and chemical processing facilities. The result is a dramatic reduction in dependence on furnace oil, LPG and natural gas, all of which India imports at considerable foreign exchange cost.

"Adoption of CASE coal gasification technologies insulates Indian industry from current geopolitical situations and global wars," Dr. Roger Kumar has stated plainly, and the precision of

As global energy markets fracture under geopolitical pressure, Dr. Roger Kumar has built CASE Group around a conviction that India's coal reserves are not a liability but its most underutilised strategic asset.

that claim is worth pausing on. India holds some of the world's largest coal reserves. Its industrial energy security challenge has never been about resource availability. It has been about conversion technology, and that is exactly the gap CASE Group has spent three decades engineering its way across.

The company's gasifier portfolio spans hot gasifiers, cold gasifiers, Circulating Fluidized Bed Coal systems and fourth-generation Entrained Flow Gasifiers, all PLC-controlled and designed to handle India's characteristically high-ash coal efficiently. The technology stack is complemented by air pollution control systems, waste heat recovery units and Zero Liquid Discharge water treatment solutions, ensuring that the transition to indigenous coal does not trade one environmental problem for another.

"By replacing imported fuels with indigenous coal, we reduce foreign exchange outflow and strengthen national energy independence," Dr. Kumar has said. "This is precisely what Atmanirbhar Bharat demands of Indian industry and of Indian engineering."

Honoured in 2026 as World's Greatest Sustainable Energy Transformation Leader by AsiaOne, Dr. Kumar's recognition is a reflection of a market beginning to understand what he argued three decades ago: that the cleanest version of coal is still coal, if the technology is right. CASE Group is that technology.